

SOURABH CHAUHAN

Software Engineer | Python Backend Developer | AI Automation & Cloud Engineer
TCS Python L3 Certified | Oracle Professional Certified GenAI
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PROFESSIONAL SUMMARY

Software Engineer with 5+ years of experience designing, developing, testing, and maintaining backend services, automation platforms, cloud-enabled tools, and enterprise web applications across regulated environments.

ACADEMIC QUALIFICATIONS

Master of Science in Computer Science
Bachelor of Computer Applications

Lakehead University, Canada | GPA: 4.0/4.0 | Aug 2024 - Sept 2025
Amity University, India | GPA 3.7/4.0 | Sept 2016 – 2019

PROFESSIONAL EXPERIENCE

AI Trainer (Python) - Outlier AI (Freelance)

Nov 2025 – Jun 2026

- Improved AI output reliability by evaluating model-generated responses against structured quality guidelines, using Python-supported validation workflows, logical reasoning checks, and specification-based review methods.
- Enhanced model accuracy and consistency across large-scale AI evaluation tasks by identifying hallucinations, reasoning gaps, formatting issues, and compliance failures in generated outputs.

Systems Engineer - Tata Consultancy Services

April 2023 – Aug 2024

Enterprise clients across healthcare, banking, insurance, and infrastructure operations

- Reduced manual IT query handling by 70% by designing and deploying an AI-assisted support platform that handled 8,000+ monthly queries using Python, NLP pipelines, REST APIs, rule-based orchestration, and structured validation logic.
- Developed a real-time diagnostics and compliance platform for server health checks, dependency validation, patch governance, and operational reporting, reducing infrastructure review effort by 95%, while improving infrastructure reliability.
- Implemented Docker-containerized microservices, Pytest-based TDD practices, and GitHub Actions CI/CD pipelines across distributed backend services, improving deployment reliability and reducing release failures.
- Partnered with client stakeholders, delivery teams, cloud teams, security teams, and infrastructure teams to gather requirements, remove technical blockers, and align automation solutions with business and operational needs.
- Improved developer and operations by standardizing reusable automation workflows, documenting support procedures, and reducing repetitive manual intervention across enterprise environments.

Asst. Systems Engineer - Tata Consultancy Services

Feb 2022 – Mar 2023

- Accelerated incident resolution by building ML-driven ticket classification and remediation workflows that improved triage accuracy, reduced manual support effort, and supported enterprise service reliability.
- Improved incident analysis and reporting accuracy by developing AWS and PySpark-based ETL pipelines that processed large operational datasets from multiple enterprise systems.
- Developed end-to-end IT automation by integrating ServiceNow, VMware vCenter, and Cisco UCS platforms for incident management, root-cause diagnostics, trigger remediation, and operational reporting.
- Modernized legacy Python and PowerShell automation workflows into modular microservices architecture with reusable deployment pipelines and scalable service boundaries.
- Delivered 12+ enterprise automation and backend solutions across the SDLC, reducing manual operational effort by 155+ hours per month.

Programmer/Automation Engineer - Tata Consultancy Services

Mar 2020 – Jan 2022

- Reduced VMware VM provisioning time from 2-3 Hours to under 10 minutes by engineering an end-to-end infrastructure automation using Python, pyVmomi, and vSphere APIs for VM provisioning, configuration, and OS hardening.
- Eliminated 98% of manual Linux applications and Modules deployment effort by building robust CI/CD deployment pipelines using Python, Ansible, Bash, shell scripting, and infrastructure automation workflows, improving release consistency across regulated enterprise environments.
- Streamlined enterprise patch management operations through SCCM-integrated Python automation scripts that improved Windows deployment throughput by 70% while maintaining compliance and security standards.
- Developed backend application for healthcare claims processing, reconciliation, and enterprise reporting systems with secure RESTful APIs, role-based access controls, and compliance-focused workflows for healthcare clients.
- Automated infrastructure maintenance workflows using Python and Bash for user provisioning, snapshot lifecycle management, log rotation, and operational support tasks, reducing repetitive workload by 80%.

- Developed core backend features for Home Services and HRMS platforms using C#, .NET, Microsoft SQL Server, and database-driven workflows for web and mobile enterprise applications.
- Implemented modules for employee access management, service booking workflows, and user-role authorization systems.
- Supported application testing, defect fixes, database updates, and technical documentation during development and release cycles.

PERSONAL PROJECTS

- **Driver Alertness Detection App** | Python, Computer Vision, TensorFlow, OpenCV | [Link](#)
 - Designed and implemented real-time facial analysis app to detect driver fatigue & impairment using behavioral cues (e.g. droopy eyes, yawning).
 - Triggered in-app alerts to prompt rest, improving road safety and reducing accident risk.
- **Video Transcript Search Engine** | Python, Whisper, Flask, NLP, Scikit-learn | [Link](#)
 - Built an AI-powered semantic video search platform that transcribed educational videos, extracted contextual keywords, and enabled topic-level search across generated transcripts using NLP pipelines and Flask APIs.
- **Ingredient Insight App** | Python, OCR, Computer Vision, Streamlit | [Link](#)
 - Developed an AI-based ingredient analysis platform that extracted product label data from images and generated health-risk insights using OCR, external nutrition datasets, and computer vision workflows.

CORE SKILLS

- **Programming:** Python, SQL, Bash, PowerShell, C/C++, Java, YAML, TypeScript, C#, .Net
- **Backend:** FastAPI, Flask, Django, REST APIs, API Integrations, Microservices
- **Cloud & DevOps:** AWS, GCP, Docker, GitHub Actions, CI/CD, Terraform, Ansible, Airflow, Linux
- **AI/ML:** Scikit-learn, NLP, TensorFlow, OpenAI API, RAG, Prompt Engineering, PyTorch
- **Databases & Data:** MySQL, MongoDB, SQLite, PySpark, Kafka, ELT Pipelines, Jupyter Notebooks, Databricks
- **Reliability & Observability:** Pytest, Unit Testing, TDD, PyATS, Selenium, Incident response, Root-cause analysis, Monitoring automations, Alerting, Logging, Power Automate, Confluent
- **Tools & Practices:** Git, Jira, Confluence, Agile/Scrum, ServiceNow, ITIL, Technical documentation, Sprint Planning

AWARDS & RECOGNITION

- **Star of the Month Award** — Tata Consultancy Services
- **On-the-Spot Team Excellence Award** for enterprise automation delivery
- **Client Recognition Awards** for improving operational efficiency across enterprise environments.
- **Special Initiative and Contextual Master Awards** for introducing AI automation and CI/CD pipeline improvements for client delivery teams

CERTIFICATIONS

- **TCS iON:** Python L3 Certified, Ansible, ITIL, Agile
- **IBM:** Data Analysis with Python | [Link](#)
- **Oracle:** Oracle Certified Generative AI Professional [Link](#) | Certified DevOps Professional [Link](#) | Certified Data Science Professional [Link](#)
- **MongoDB:** MongoDB for SQL Experts | [Link](#)